

Bill Of Materials M001

Part Name/Description	Unit Quantity	Unit Prototype Cost	Total Prototype Cost	Unit Production Cost	Total Production Cost	Manufacturer	Manufacturer Part #	Vendor Link
ESP 32 - Wroom N8	1	\$5.49	\$5.49	\$5.49	\$5.49	Espressif	ESP32-S3-WROOM-1-N	https://www.digikey.com
CAP CER 330PF 16V X7R	2	\$0.10	\$0.20	\$0.10	\$0.20	Murata	GRM033R71C331KA01	https://www.digikey.com
CAP CER 10000PF 10V X	2	\$0.08	\$0.16	\$0.08	\$0.16	Murata	GRM022C71A103KE01	https://www.digikey.com
CAP CER 0.022UF 6.3V X	1	\$0.10	\$0.10	\$0.10	\$0.10	Murata	GRM033R60J223KE01	https://www.digikey.com
CAP 0805 250V 33 NF X7	6	\$0.18	\$1.08	\$0.18	\$1.08	Yageo	CC0805KKX7RYBB333	https://www.digikey.com
CAP CER 0.1UF 10V X5R	7	\$0.08	\$0.56	\$0.08	\$0.56	Murata	GRM033R61A104KE15	https://www.digikey.com
CAP CER 0.47UF 6.3V X5	1	\$0.08	\$0.08	\$0.08	\$0.08	Murata	GRM033R60J474KE90	https://www.digikey.com
CAP CER 2.2UF 16V X5R	2	\$0.10	\$0.20	\$0.10	\$0.20	Samsung Electr	CL10A225KO8NNNC	https://www.digikey.com
CAP CER 10UF 16V X5R	1	\$0.08	\$0.08	\$0.08	\$0.08	Taiyo Yuden	EMK212BJ106KG-T	https://www.digikey.com
CAP CER 22UF 6.3V 20%	4	\$0.28	\$1.12	\$0.28	\$1.12	Murata	GRM155R60J226ME11	https://www.digikey.com
CAP CER 100UF 16V X5R	5	\$1.74	\$8.70	\$1.74	\$8.70	Taiyo Yuden	EMK325ABJ107MM-T	https://www.digikey.com
CAP CER 150UF 6.3V X5R	1	\$1.31	\$1.31	\$1.31	\$1.31	Samsung Electr	CL32A157MQVNNNE	https://www.digikey.com
RES 10 OHM 5% 2W 2512	1	\$0.44	\$0.44	\$0.44	\$0.44	Bourns	CRM2512-JW-100ELF	https://www.digikey.com
RES 20 OHM 5% 2W 2512	1	\$0.24	\$0.24	\$0.24	\$0.24	Yageo	AC2512JK-7W20RL	https://www.digikey.com
RES 220 OHM 5% 2W 251	1	\$0.53	\$0.53	\$0.53	\$0.53	TE Connectivity	CRGH2512J220R	https://www.digikey.com
RES 1.65K OHM 1% 2W 1	1	\$1.39	\$1.39	\$1.39	\$1.39	TE Connectivity	3503G2B1K65FTD	https://www.digikey.com
RES 5.11K OHM 1% 2W 2	1	\$0.52	\$0.52	\$0.52	\$0.52	Bourns	CRM2512-FX-5111ELF	https://www.digikey.com
RES 10K OHM 1% 2W 25	1	\$0.51	\$0.51	\$0.51	\$0.51	TE Connectivity	CRGP2512F10K	https://www.digikey.com
RES 49.9K OHM 1% 2W 2	1	\$0.52	\$0.52	\$0.52	\$0.52	Bourns	CRM2512-FX-4992ELF	https://www.digikey.com
RES 62K OHM 5% 1/8W 0	1	\$10.00	\$10.00	\$10.00	\$10.00	Yageo	RC0805JR-0762KL	https://www.digikey.com
RES SMD 196K OHM 1%	1	\$0.10	\$0.10	\$0.10	\$0.10	Vishay Dale	RES SMD 196K OHM 1	https://www.digikey.com
DIODE SCHOTTKY 100V	3	\$0.40	\$1.20	\$0.40	\$1.20	Vishay	VS-30WQ10FNTR-M3	https://www.digikey.com
LED GREEN CLEAR SMD	1	\$0.10	\$0.10	\$0.10	\$0.10	Lite-On	LTST-C170GKT	https://www.digikey.com
FUSE GLASS 5A 250VAC	1	\$0.63	\$0.63	\$0.63	\$0.63	Schurter	34.3124	https://www.digikey.com
FUSE GLASS 4A 250VAC	1	\$0.63	\$0.63	\$0.63	\$0.63	Schurter	34.3123	https://www.digikey.com
FIXED IND 4.7UH 2.5A 14	1	\$0.19	\$0.19	\$0.19	\$0.19	Taiyo Yuden	NRS4018T6R8MDGJ	https://www.digikey.com
FIXED IND 33UH 2.5A 93	1	\$1.06	\$1.06	\$1.06	\$1.06	Murata	1274AS-H-330M=P3	https://www.digikey.com
Positional Rotation DC Mo	2	\$5.95	\$11.90	\$5.95	\$11.90	Adafruit	4326	https://www.digikey.com
STANDARD MOTOR 1160	3	\$3.71	\$11.13	\$3.71	\$11.13	NMB	PPN7PA12C1	https://www.digikey.com
8 Pin Connector	2	\$1.48	\$2.96	\$1.48	\$2.96	Molex	702460801	https://www.digikey.com
8 Pin Ribbon Cable	2	\$5.00	\$10.00	\$5.00	\$10.00	Amphenol	10124667-Z0025YYLF	https://www.digikey.com
Power Barrel Connector Ja	1	\$0.71	\$0.71	\$0.71	\$0.71	Same Sky	PJ-102AH	https://www.digikey.com
USB - micro B USB 2.0	1	\$0.77	\$0.77	\$0.77	\$0.77	GCT	USB3131-30-0230-A	https://www.digikey.com
Switch button	2	\$0.10	\$0.20	\$0.10	\$0.20	Same Sky	TS02-66-55-BK-100-LC	https://www.digikey.com
SWITCH PB DPDT 0.1A 3	1	\$0.46	\$0.46	\$0.46	\$0.46	E-Switch	TL2233OA	https://www.digikey.com
IC HALF BRIDGE DRIVER	3	\$3.55	\$10.65	\$3.55	\$10.65	Infineon	IFX9201SGAUMA1	https://www.digikey.com

Buck Switching Regulator	1	\$3.77	\$3.77	\$3.77	\$3.77	Texas Instrume	LM5005MHX/NOPB	https://www.digikey.com/product-detail/en/texas-instruments/lm5005mhx/nopb/296-1447-1
Buck Switching Regulator	1	\$0.71	\$0.71	\$0.71	\$0.71	Diode Incorpora	AP63203WU-7	https://www.digikey.com/product-detail/en/diodes-incorporated/ap63203wu-7/296-1447-1
3 Pin Screw Terminal	2	\$0.71	\$1.42	\$0.71	\$1.42	Adam Tech	EBWA-03-A	https://www.digikey.com/product-detail/en/adam-tech/ebwa-03-a/296-1447-1
2 Pin Screw Terminal	3	\$0.32	\$0.96	\$0.32	\$0.96	Same Sky	TB002-500-02BE	https://www.digikey.com/product-detail/en/same-sky/tb002-500-02be/296-1447-1
		Total Proto	\$87.09	Total Product	\$87.09			

JI Subsystem

Datasheet Link	Supplier	Supplier Part #	# Ordered	Date Ordered	# Received	Surplu s	Schematic Reference Designators
https://documentation.murata.com/Products/Passive%20Components/Resistors/Thin%20Film/0402/0402%20Thin%20Film%20Resistors/5407-ESP32-S	Digikey	5407-ESP32-S	3	3/12/2026	3	2	U4
https://search.murata.com/Products/Passive%20Components/Resistors/Thin%20Film/0402/0402%20Thin%20Film%20Resistors/490-11359-1-ND	Digikey	490-11359-1-ND	2	3/12/2026	2	0	C5,C10
https://search.murata.com/Products/Passive%20Components/Resistors/Thin%20Film/0402/0402%20Thin%20Film%20Resistors/490-GRM022C	Digikey	490-GRM022C	1	3/12/2026	2	0	C1,C7
https://search.murata.com/Products/Passive%20Components/Resistors/Thin%20Film/0402/0402%20Thin%20Film%20Resistors/490-3169-1-ND	Digikey	490-3169-1-ND	1	3/12/2026	1	0	C8
https://www.yageo.com/Products/Passive%20Components/Resistors/Thin%20Film/0402/0402%20Thin%20Film%20Resistors/13-CC0805KK	Digikey	13-CC0805KK	1	3/12/2026	6	0	C23,C24,C25,C26,C27,C28
https://search.murata.com/Products/Passive%20Components/Resistors/Thin%20Film/0402/0402%20Thin%20Film%20Resistors/490-5881-1-ND	Digikey	490-5881-1-ND	10	3/12/2026	10	3	C11,C16,C18,C21,C22,C30,C32
https://search.murata.com/Products/Passive%20Components/Resistors/Thin%20Film/0402/0402%20Thin%20Film%20Resistors/490-10409-1-ND	Digikey	490-10409-1-ND	1	3/12/2026	1	0	C4
https://mm.digikey.com/ViewSiteDetail.aspx?lang=en/resistor/part/1276-1040-1-ND	Digikey	1276-1040-1-ND	3	3/12/2026	3	1	C2,C3
https://mm.digikey.com/ViewSiteDetail.aspx?lang=en/resistor/part/587-1295-1-ND	Digikey	587-1295-1-ND	1	3/12/2026	1	0	C6
https://search.murata.com/Products/Passive%20Components/Resistors/Thin%20Film/0402/0402%20Thin%20Film%20Resistors/490-GRM155R	Digikey	490-GRM155R	6	3/12/2026	6	2	C9,C12,C14,C15
https://mm.digikey.com/ViewSiteDetail.aspx?lang=en/resistor/part/587-3152-1-ND	Digikey	587-3152-1-ND	7	3/12/2026	7	2	C17,C19,C20,C29,C31
https://weblib.samsongroup.com/Products/Passive%20Components/Resistors/Thin%20Film/0402/0402%20Thin%20Film%20Resistors/1276-3367-1-ND	Digikey	1276-3367-1-ND	1	3/12/2026	1	0	C13
https://www.bourns.com/Products/Passive%20Components/Resistors/Thin%20Film/0402/0402%20Thin%20Film%20Resistors/CRM2512-JW	Digikey	CRM2512-JW	1	3/12/2026	1	0	R5
https://yageogroup.com/Products/Passive%20Components/Resistors/Thin%20Film/0402/0402%20Thin%20Film%20Resistors/13-AC2512JK	Digikey	13-AC2512JK	1	3/12/2026	1	0	R2
https://www.te.com/Products/Passive%20Components/Resistors/Thin%20Film/0402/0402%20Thin%20Film%20Resistors/A126460CT-NI	Digikey	A126460CT-NI	1	3/12/2026	1	0	R9
https://www.te.com/Products/Passive%20Components/Resistors/Thin%20Film/0402/0402%20Thin%20Film%20Resistors/1712-3503G2B	Digikey	1712-3503G2B	1	3/12/2026	1	0	R3
https://www.bourns.com/Products/Passive%20Components/Resistors/Thin%20Film/0402/0402%20Thin%20Film%20Resistors/118-CRM2512	Digikey	118-CRM2512	1	3/12/2026	1	0	R4
https://www.te.com/Products/Passive%20Components/Resistors/Thin%20Film/0402/0402%20Thin%20Film%20Resistors/A130733CT-NI	Digikey	A130733CT-NI	1	3/12/2026	1	0	R6,R7,R10,R11
https://www.bourns.com/Products/Passive%20Components/Resistors/Thin%20Film/0402/0402%20Thin%20Film%20Resistors/118-CRM2512	Digikey	118-CRM2512	1	3/12/2026	1	0	R1
https://yageogroup.com/Products/Passive%20Components/Resistors/Thin%20Film/0402/0402%20Thin%20Film%20Resistors/311-62KARDK	Digikey	311-62KARDK	0	3/12/2026	1	0	R8
https://www.vishay.com/Products/Passive%20Components/Resistors/Thin%20Film/0402/0402%20Thin%20Film%20Resistors/541-196KCCT	Digikey	541-196KCCT	0	3/12/2026	1	0	R12
https://www.vishay.com/Products/Passive%20Components/Resistors/Thin%20Film/0402/0402%20Thin%20Film%20Resistors/112-VS-30WQ	Digikey	112-VS-30WQ	7	3/12/2026	7	4	D1,D2,D3,D4,D6
https://mm.digikey.com/ViewSiteDetail.aspx?lang=en/resistor/part/160-1179-1-ND	Digikey	160-1179-1-ND	1	3/12/2026	1	0	D5
https://www.digikey.com/Products/Passive%20Components/Resistors/Thin%20Film/0402/0402%20Thin%20Film%20Resistors/486-1223-ND	Digikey	486-1223-ND	0	3/12/2026	1	0	F1
https://www.schurter.com/Products/Passive%20Components/Resistors/Thin%20Film/0402/0402%20Thin%20Film%20Resistors/486-1222-ND	Digikey	486-1222-ND	5	3/12/2026	5	4	F2
https://mm.digikey.com/ViewSiteDetail.aspx?lang=en/resistor/part/587-2894-6-ND	Digikey	587-2894-6-ND	0	3/12/2026	1	0	L2
https://search.murata.com/Products/Passive%20Components/Resistors/Thin%20Film/0402/0402%20Thin%20Film%20Resistors/490-10851-1-ND	Digikey	490-10851-1-ND	1	3/12/2026	1	0	L1
https://mm.digikey.com/ViewSiteDetail.aspx?lang=en/resistor/part/1528-4326-ND	Digikey	1528-4326-ND	3	3/12/2026	3	1	M1,M2
https://mm.digikey.com/ViewSiteDetail.aspx?lang=en/resistor/part/942-PPN7PA1	Digikey	942-PPN7PA1	3	3/12/2026	3	0	M3, M4, M5
https://www.molex.com/Products/Passive%20Components/Resistors/Thin%20Film/0402/0402%20Thin%20Film%20Resistors/900-07024608	Digikey	900-07024608	0	3/12/2026	2	0	J3, J4
https://cdn.amphenol.com/Products/Passive%20Components/Resistors/Thin%20Film/0402/0402%20Thin%20Film%20Resistors/10124667-Z00	Digikey	10124667-Z00	0	3/12/2026	2	0	J3, J4
https://www.cuidevi.com/Products/Passive%20Components/Resistors/Thin%20Film/0402/0402%20Thin%20Film%20Resistors/CP-102AH-ND	Digikey	CP-102AH-ND	1	3/12/2026	1	0	J1
https://gct.co/files/schematic/2073-USB3131	Digikey	2073-USB3131	0	3/12/2026	1	0	J2
https://www.samesky.com/Products/Passive%20Components/Resistors/Thin%20Film/0402/0402%20Thin%20Film%20Resistors/2223-TS02-66	Digikey	2223-TS02-66	0	3/12/2026	2	0	SW1,SW2
https://configured-pcb.com/Products/Passive%20Components/Resistors/Thin%20Film/0402/0402%20Thin%20Film%20Resistors/141-TL2233OA	Digikey	141-TL2233OA	1	3/12/2026	1	0	S1
https://www.infineon.com/Products/Passive%20Components/Resistors/Thin%20Film/0402/0402%20Thin%20Film%20Resistors/IFX9201SGAUMA1CT-NI	Digikey	IFX9201SGAUMA1CT-NI		3/12/2026	3	0	U1,U2,U3

https://www.ti.com/	Digikey	LM5005MHX/N	2	3/12/2026	2	1	VR1
https://www.diodes.com/	Digikey	AP63203WU-7	2	3/12/2026	2	1	VR2
https://app.adam-tech.com/	Digikey	2057-EBWA-03	2	3/12/2026	2	0	M1,M2
https://www.cuidevice.com/	Digikey	102-6145-ND	0	3/12/2026	3	0	M3, M4, M5